

FULL STACK ENGINEERING PROJECT REPORT

On

Archon-AI

Submitted in partial fulfilment of the requirement for the

Course Full Stack Engineering (22CS037) of

COMPUTER SCIENCE AND ENGINEERING

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2310991326

CERTIFICATE

This is to be certified that the project entitled “**Archon-AI**” has been submitted for the Bachelor of Computer Science Engineering at Chitkara University, Punjab during the academic semester January 2026 - May 2026. It is a bonafide piece of project work carried out by **Yuvraj Gupta, (2310991326)** towards the partial fulfillment for the award of the course **Full Stack Engineering (22CS037)** under the guidance and supervision of **Dr. Suhasini**.

Sign. of Project Guide:

Dr. Suhasini

Assistant Professor

CSE, CUIET

CANDIDATE'S DECLARATION

I, **Yuvraj Gupta (2310991326)**, B.E.-2023 of the Chitkara University, Punjab hereby declare that the Full Stack Engineering (22CS037) Project Report entitled “**ARCHON-AI**” is an original work and data provided in the study is authentic to the best of my knowledge. This report has not been submitted to any other Institute for the award of any other course.

Sign. of Student 1

Yuvraj Gupta

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Place: Chitkara University, Rajpura, Punjab

Date: 23/05/2026

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Lastly, I would like to thank the almighty and my parents for their moral support and friends with whom we shared our day-to day experience and received lots of suggestions that improve our quality of work.

Yuvraj Gupta

2310991326

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1. ABSTRACT

Archon-AI is an intelligent, multi-tenant enterprise AI orchestration and learning ecosystem designed to consolidate fragmented digital workspaces into a unified portal. In modern computing and educational fields, users are constantly overwhelmed by a disconnected sea of productivity tools, separate AI chat dashboards, and isolated communication hubs. This platform resolves these structural fragmentations by combining advanced workspace customization with cutting-edge AI features natively into a single ecosystem.

Built entirely using the MERN stack (MongoDB, Express.js, React.js, and Node.js), Archon-AI features an interactive central engine powered by the Google Gemini API. This native integration allows the app to process natural language queries, deliver real-time intelligent chatbot responses, analyze unstructured documentation, and dynamically generate educational tools such as automated quizzes and revision flashcards.

Furthermore, the platform incorporates a robust multi-tiered authentication setup, real-time collaboration rooms, a collaborative workspace whiteboard, and modular configuration dashboards. By marrying AI capabilities with structured utility modules, Archon-AI elevates workflow efficiency, enhances knowledge retention, and redefines how developers, students, and enterprise users approach digital productivity.

Keywords: Archon-AI, MERN Stack, Google Gemini API, Multi-Tenant Workspace, Enterprise Automation, AI Chatbot, Interactive Whiteboard, Intelligent Learning Systems.

2. INTRODUCTION TO THE PROJECT

In the modern digital era, enterprise teams, students, and software developers interact daily with a vast array of web platforms to complete simple workflows. The distribution of data across completely independent environments—such as dedicated AI chat agents, note-taking canvas apps, team communication software, and independent file systems—creates severe cognitive overhead. Valuable attention spans are lost to perpetual browser context-switching and fragmented data streams.

To reverse these structural limits, there is a crucial demand for comprehensive platforms capable of bringing both general productivity software and generative artificial intelligence engines under a single unified dashboard.

Archon-AI directly tackles this issue by deploying an advanced web platform where users build, organize, and manage deep workspaces embedded with native AI automation. Rather than operating as an isolated utility, Archon-AI unifies its tools. For example, text generated inside the AI chat interface can be moved directly into collaborative whiteboards or cross-referenced with documentation libraries.

Leveraging the Google Gemini API, the portal serves as a powerful backend processor to parse complicated documents, summarize text blocks, and instantly turn long raw files into interactive learning materials such as adaptive quizzes or structured review decks. Developed on the scalable MERN stack architecture, Archon-AI provides modern, secure, and fast data retrieval loops alongside highly responsive multi-user experiences.

2.1 Background

Historically, application designs treated text editing, team chat, and data engines as totally separate categories. When generative AI broke out, it was primarily introduced as basic standalone chat windows. This design forced users to manually copy information back and forth between their workspace docs and external AI windows.

As standard enterprise operations move fast, this back-and-forth approach exposes organizations to data leaks, slows down project shipping speeds, and breaks critical visibility for team managers.

By utilizing modern database structures, secure token protocols, and reliable real-time server connections, we can engineer software architectures that place AI assistants right inside the collaborative workspace. Archon-AI was built from the ground up on this principle: delivering an optimized environment where file tracking, real-time whiteboards, and intelligent agents communicate instantly on a unified system.

2.2 Problem Statement

The current digital ecosystem presents distinct flaws across personal organization and corporate workflow management:

- **Context-Switching Overhead:** Users run too many independent tools simultaneously for document browsing, communication, drawing, and AI prompting, which decreases focus and lowers overall work output.
- **Static Document Formats:** Standard document repositories lack deep search tools, interactive explanation mechanisms, and automated study-aid builders, leaving users stuck with slow, manual reading steps.
- **Disconnected Creative Frameworks:** Traditional digital drawing and engineering design spaces do not connect with backend generative models, blocking the path from initial ideation to structured code.
- **Weak Data Consolidation:** Outdated tracking systems struggle to display real-time user metrics, quiz score trends, or project task progression securely inside a cohesive cockpit view.

Archon-AI solves these points by providing a secure, centralized, multi-tenant workspace architecture that blends real-time communication tools, multi-media whiteboards, and robust AI automation into one platform.



3. SOFTWARE AND HARDWARE REQUIREMENT SPECIFICATION

Building a complex, real-time AI ecosystem requires an aligned suite of development frameworks, deployment microservices, and targeted hardware setups. Archon-AI utilizes a full-stack JavaScript architecture designed to minimize latency during heavy AI streaming and real-time canvas rendering operations.

3.1 Methods

The development and architectural planning of Archon-AI follow a highly structured, component-driven Full-Stack Software Engineering Lifecycle. Instead of relying on monolithic design patterns, the platform implements a modular Model-View-Controller (MVC) decoupling paradigm through the MERN tech stack. This separation ensures that high-volume data streaming, complex browser canvas updates, and heavy AI processing workflows can operate independently without creating performance bottlenecks.

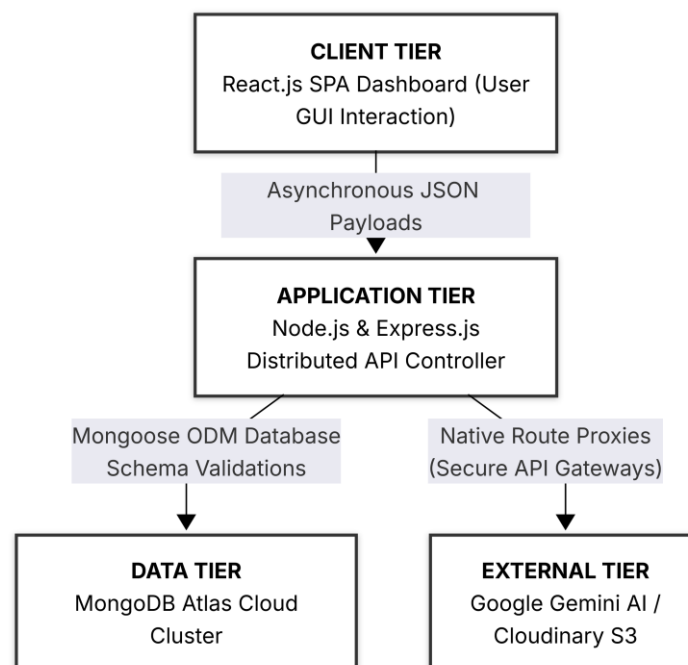


Fig 3.1.1: The System Workflow

The system workflow is broken down into five core procedural phases designed to provide a highly performant user experience:

- **Secure Client Ingestion Phase:** The user authenticates securely via JSON Web Tokens (JWT) stored safely inside HTTP-only cookies . Once inside the personalized dashboard, the client-side single-page application (SPA) initializes active listeners for both real-time drawing actions and asynchronous AI query operations .
- **Document Processing & Vectorization Pipeline:** When a user uploads an unstructured PDF through the dashboard, the binary asset is securely streamed to the Express backend. The server distributes a copy of the binary file to Cloudinary for cloud hosting and reads the text stream locally to clean the raw characters.
- **Generative Orchestration Loop:** Once the system cleans the text context from the document, it anchors the text dynamically within a Mongoose session wrapper. When a user calls a specific AI utility (e.g., summary generation or quiz creation), the application server wraps the context text inside a targeted structural template before dispatching it to the Google Gemini API .
- **Bidirectional Synchronization Layer:** For collaborative canvas areas (the project whiteboards), the platform uses a state-driven socket listener. Every mouse movement, stroke creation, or canvas transformation is converted into string coordinate objects and broadcast to all connected team members instantly.
- **Persistent Analytics Capture:** Every operational state change—including test score results, total chat loops completed, and newly created revision card decks—is routed through strict validation logic before being safely committed to the cloud database cluster.

3.2 Programming / Working Environment

To ensure a highly uniform engineering pipeline and support frictionless continuous deployment, the development environment for Archon-AI uses the following language runtimes, libraries, and tools:

Frontend Engineering Suite

- **React.js (v18.x):** Utilized as the fundamental single-page view architecture. It leverages functional components, advanced Hooks (useState, useEffect, useRef, useContext), and Virtual DOM comparisons to handle complex frontend rendering dynamically without forced browser refreshes.
- **Axios HTTP Client:** Used to handle asynchronous promise-based connection loops with backend route endpoints, featuring centralized interceptors that capture server responses and gracefully isolate communication failures.
- **HTML5 Canvas API:** The core low-level browser technology used to power the real-time whiteboard module, enabling responsive pixel-level tracking, path calculations, and rapid redraw loops.

Backend Frameworks & Runtimes

- **Node.js (LTS v20.x):** Serves as the high-performance V8 JavaScript execution runtime on the server side, utilizing its non-blocking, event-driven I/O loop to process thousands of concurrent user actions efficiently.
- **Express.js (v4.19):** A minimal, flexible web framework deployed on top of Node.js to configure clear RESTful API routes, manage custom middleware processing loops, and regulate incoming user data payloads.
- **JSON Web Token (JWT):** The stateless cryptography protocol chosen to implement access authentication, issuing secure, cryptographically signed cookies to safeguard user identity states across sessions.

Database & External Core Gateways

- **MongoDB Atlas Cluster:** Deployed as the primary non-relational database engine. It stores transactional document histories, analytical states, and platform metadata inside dynamic BSON collections.
- **Mongoose ODM:** Used as the structural Object Data Modeling library for Node.js, establishing explicit server-side validation properties, auto-casting data fields, and simplifying complex data retrieval paths.

- **Google Gemini API SDK:** The advanced foundational AI framework integrated directly into our core application controllers to analyze long text strings, handle smart chat messaging, and compile formatted quiz outputs.
- **Cloudinary Node SDK:** Integrated to manage file uploads securely over HTTPS, offloading heavy binary storage burdens from the primary application server via an automated cloud media pipeline.

Developer Utility Frameworks

- **Visual Studio Code:** The primary Integrated Development Environment (IDE), augmented with syntax validators, formatting utilities, and real-time terminal tracking screens.
- **Postman API Suite:** Used to execute systematic route verification testing, trace JSON payload formats, validate request response times, and stress-test endpoint rate limits.
- **Git & GitHub Platform:** Deployed as the core version control ecosystem to maintain clean code branching, review implementation pull requests, and track code revisions securely

3.3 Requirements to Run the Application

3.3.1 Hardware Specifications

The application architecture is optimized to deploy effortlessly on standard computing setups without demanding specialized graphics processing units or enterprise-grade server infrastructure:

- **Central Processing Unit (CPU):** A multi-core modern processor architecture (Intel Core i5 10th Gen, AMD Ryzen 5, Apple Silicon M-Series, or equivalent) running at a base clock speed of 2.4GHz. This ensures the backend can execute real-time serialization tasks smoothly.
- **Random Access Memory (RAM):** A minimum of 8GB of physical memory is required for development, allowing the concurrent operation of local Node.js servers,

active browser debugging instances, and local code packaging tools. A minimum of 4GB is required for basic client execution.

- **Storage Capacity:** A minimum of 5GB of unallocated solid-state drive (SSD) space to host project dependencies, caching libraries, and system logging directories.
- **Network Adapter Interface:** A standard IEEE 802.11ac Wi-Fi or Gigabit Ethernet card capable of maintaining a stable internet connection with a bandwidth of >10Mbps to handle external cloud API queries smoothly.

Component	Specification
CPU	Multi-core processor (Intel i5 10th Gen / Ryzen 5 / Apple M-Series) @ ≥ 2.4 GHz base clock.
RAM	8GB minimum for development (local servers/debugging); 4GB minimum for client execution.
Storage	≥ 5 GB available Solid-State Drive (SSD) space for dependencies and logs.
Network	IEEE 802.11ac Wi-Fi or Gigabit Ethernet with >10Mbps bandwidth.

Table 3.3.1: Hardware Requirements

3.3.2 Software Specifications

To maintain complete platform cross-compatibility across diverse consumer ecosystems, the platform complies with the following runtime standards:

- **Operating System Environment:** Fully certified across Microsoft Windows 10/11 (64-bit builds), Apple macOS (Monterey or later), and mainline Linux distributions like Ubuntu 22.04 LTS.
- **Client Web Browser Interface:** Optimized for modern engines equipped with advanced V8 interpretation environments, including Google Chrome (v115+), Microsoft Edge (v115+), Mozilla Firefox (v118+), and Apple Safari (v17+).
- **System Runtime Environments:** Node.js runtime installation matching the long-term support release standard (LTS v20.x or later) along with the Node Package Manager (npm v10.x+).



4. DATABASE ANALYZING, DESIGN AND IMPLEMENTATION

Archon-AI uses MongoDB as its primary database because its flexible document structure makes it highly scalable and capable of handling semi-structured data efficiently. Traditional relational database setups fall short here because AI chat histories, dynamic whiteboard drawings, and randomly generated quiz formats vary too much in size and shape to fit into rigid tables with fixed columns.

By leveraging an agile BSON document-store architecture, Archon-AI keeps its schema flexible without sacrificing data speed. The database framework uses embedded document trees for fast internal operations, but separates larger data models with clean structural links where necessary to prevent documents from hitting MongoDB's 16MB file size ceiling.

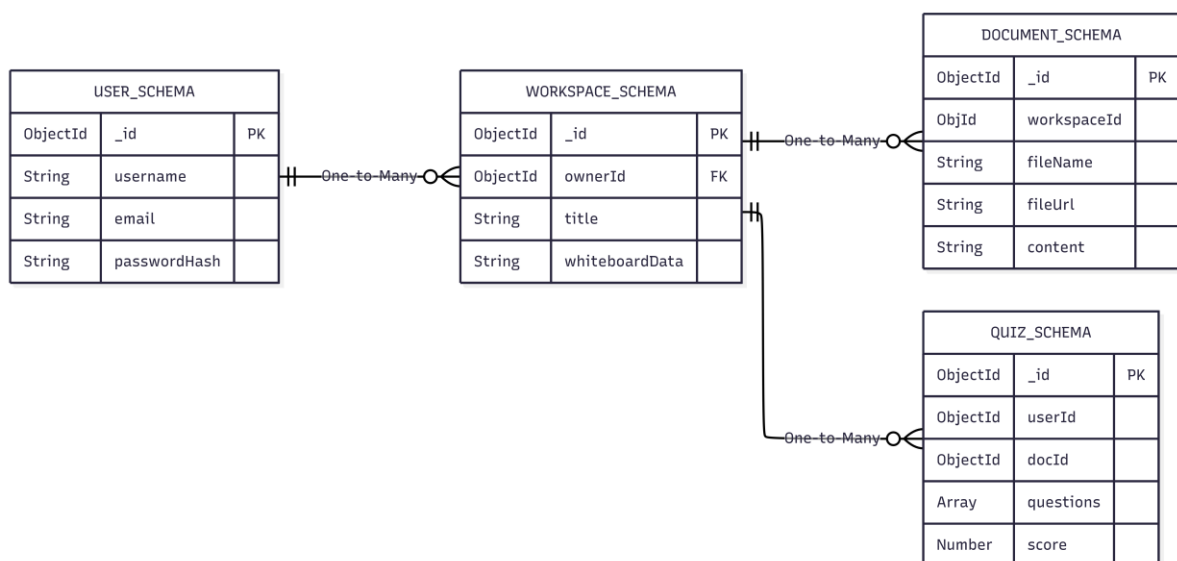


Fig 4.1: User Schema Design

4.1 Detailed Schema Implementation and Field Mappings

The Object Data Modeling layer of our server backend is written explicitly via Mongoose compilation definitions. Below is the comprehensive architectural data field structural reference used within the platform code layout:

4.1.1 User Document Schema

The core user model maps administrative identity states, cryptographic security flags, and primary account metadata :

- `_id` (mongoose.Schema.Types.ObjectId): The unique, auto-indexed primary key identifier generated by MongoDB for every new user.
- `username` (String): The unique screen identity chosen by the user, trimmed of whitespace, forced into lower-case encoding, and index-locked to prevent duplicates.
- `email` (String): The validated communication address, checked against explicit regex patterns to ensure proper email syntax.
- `password` (String): A secure string hash generated using cryptographically strong hashing algorithms to ensure sensitive user keys are never stored in plaintext format.
- `createdAt` (Date): An auto-generated timestamp tracking the exact millisecond the user account was initialized.

4.1.2 Workspace Tenant Document Schema

The workspace model tracks collaborative environments, matching explicit workspace instances back to an owner:

- `_id` (mongoose.Schema.Types.ObjectId): The primary lookup key for the workspace tenant.
- `ownerId` (mongoose.Schema.Types.ObjectId): A structural foreign key field that connects the workspace directly to the User collection.
- `title` (String): A descriptive name for the workspace, defaulted to "Untitled Workspace" if left empty by the creator.
- `collaborators` (Array): A list containing multiple user identification keys, granting collaborative read and write access to other registered users.

- `whiteboardData` (String): A high-capacity, elastic text field that saves raw canvas drawing paths, strokes, and vector coordinates as compressed JSON strings.

4.1.3 Document Repository Schema

This model tracks metadata and raw text extracted from uploaded workspace documents:

- `_id` (mongoose.Schema.Types.ObjectId): The unique reference key for the file entry.
- `workspaceId` (mongoose.Schema.Types.ObjectId): A critical foreign key mapping that binds the file directly to its parent workspace environment.
- `fileName` (String): The original file name uploaded by the user.
- `fileUrl` (String): The verified remote URL provided by Cloudinary, pointing to where the hosted asset can be accessed over secure HTTPS links.
- `fileSize` (Number): The size of the file measured in bytes, used to enforce storage caps and display asset metrics.
- `extractedContent` (String): A large text storage field containing the raw text parsed from the uploaded document, which serves as the core context source for downstream AI calls.

4.1.4 Conversational Chat Schema

Tracks historical multi-turn text conversations with integrated AI assistants:

- `_id` (mongoose.Schema.Types.ObjectId): The primary tracking identifier for a chat session.
- `userId` (mongoose.Schema.Types.ObjectId): Maps the conversation directly to its owner.
- `workspaceId` (mongoose.Schema.Types.ObjectId): Binds the conversation to a specific workspace workspace.
- `messages` (Array): An embedded subdocument array structured as follows:
 - `sender` (String): Must be set explicitly to either "user" or "model" .
 - `text` (String): The raw text of the message .
 - `timestamp` (Date): The exact execution time of the message.

4.1.5 Evaluation Quiz Schema

Stores automatically compiled test sets and matches user score histories:

- `_id` (mongoose.Schema.Types.ObjectId): The unique primary key tracking the quiz instance.
- `userId` (mongoose.Schema.Types.ObjectId): The structural user link tracking who attempted the test.
- `documentId` (mongoose.Schema.Types.ObjectId): The document lookup link showing which file was used to generate the quiz.
- `questions` (Array): An array of structured questions containing:
 - `questionText` (String): The actual question generated by the AI.
 - `options` (Array of String): A list of possible multiple-choice answers.
 - `correctAnswer` (String): The correct choice text.
- `userScore` (Number): The final calculation tracking how well the user scored, ranging from \$0\$ to \$100\$

Schema / Collection	Core Identifiers & Keys	Main Data Fields & Types	Relationships / Connections
4.1.1 User	<code>_id</code> (ObjectId)	<code>username</code> (String), <code>email</code> (String), <code>password</code> (String), <code>createdAt</code> (Date)	Primary account identity baseline.
4.1.2 Workspace	<code>_id</code> (ObjectId)	<code>title</code> (String), <code>whiteboardData</code> (JSON String), <code>collaborators</code> (Array)	<code>ownerId</code> → Links to User collection.
4.1.3 Document	<code>_id</code> (ObjectId)	<code>fileName</code> (String), <code>fileUrl</code> (String), <code>fileSize</code> (Number), <code>extractedContent</code> (String)	<code>workspaceId</code> → Links to Workspace.
4.1.4 Chat	<code>_id</code> (ObjectId)	<code>messages</code> (Array of subdocuments: <code>sender</code> , <code>text</code> , <code>timestamp</code>)	<code>userId</code> → User <code>workspaceId</code> → Workspace.
4.1.5 Quiz	<code>_id</code> (ObjectId)	<code>questions</code> (Array of: <code>questionText</code> , <code>options</code> , <code>correctAnswer</code>), <code>userScore</code> (Number)	<code>userId</code> → User <code>documentId</code> → Document.

Table 4.1: Database Schema

5. PROGRAM'S STRUCTURE ANALYZING AND GUI CONSTRUCTING

5.1 Comprehensive System Module Breakdown

The structural execution environment of Archon-AI is organized around five primary decoupled service blocks:

- **Cryptographic Identity Guard Engine:** This module handles user onboarding and secures session access pathways . It processes user registrations, securely hashes passwords on the backend, and issues encrypted JSON Web Tokens (JWT) inside protected browser cookies to prevent unauthorized API requests.
- **Dynamic Workspace Cockpit Controller:** The primary control room of the frontend application . It orchestrates workspace routing states, dynamically displays current analytics cards (such as files processed or total flashcards mastered), and updates system activity histories in real time .
- **Asynchronous AI Orchestration Gateway:** This critical connection pipeline translates custom user commands into structured requests for our generative models. It retrieves historical message feeds, passes text extracts to our endpoints as background knowledge, and streams intelligent text outputs back to the client UI .
- **Vector-Driven Interactive Whiteboard Subsystem:** A creative canvas module built using HTML5 . It tracks user pointer actions, calculates precise drawing paths, and uses web socket networks to broadcast drawing updates to all active workspace collaborators instantly.
- **Intelligent Evaluation & Assessment Builder:** This subsystem automatically parses long document data to generate interactive learning aids. It converts blocks of raw text into structured arrays of multiple-choice questions or multi-sided revision flashcards, helping users test their knowledge retention .

5.2 Graphical User Interface (GUI) Execution Mapping

(Note: Use this section to map your interface design states. Insert your screenshots directly below these reference markers to show the screen flow) .

- **Fig 5.1: Access Portal Page** – A minimal entry layout containing login and signup forms, complete with instant frontend field validation alerts.
- **Fig 5.2: Central Command Cockpit** – The main system hub, displaying core performance counters and tracking global workspace activity histories.
- **Fig 5.3: Workspace Document Explorer** – A clean layout for tracking uploaded text documents, indicating file sizes and showing attached study aids.
- **Fig 5.4: Split-View Document Analysis Panel** – Places the core document reader directly alongside quick action triggers for deep content analysis.
- **Fig 5.5: Embedded AI Context Chat Interface** – An interactive chat engine that provides real-time answers based on open document topics.
- **Fig 5.6: Automation Portal Section** – A dedicated control panel containing immediate shortcuts to build long summaries or break down specialized concepts.
- **Fig 5.7: Flashcard Set Deck Manager** – Organizes generated card arrays, timestamping creation dates and counting active cards.
- **Fig 5.8: Active Revision View** – An interactive card-flipping window built for self-paced learning and knowledge recall.
- **Fig 5.9: Examination Overview Center** – Lists available automated tests, showing question counts and saving previous scores.
- **Fig 5.10: Quiz Evaluation Screen** – A clean testing window featuring dynamic progress bars and selectable question options.
- **Fig 5.11: Performance Analytics Form** – Delivers a detailed breakdown of quiz results, highlighting total scores and listing correct answers.
- **Fig 5.12: Unified Study-Deck Analytics Dashboard** – Provides a bird's-eye view of learning metrics across all generated card decks.
- **Fig 5.13: Profile Settings Configuration Panel** – A dedicated security view allowing users to update their identity data or refresh passwords.



Create an Account

Get started with Archon AI

Full Name

Email

Password

[Create Account](#)

Already have an account? [Sign In](#)

Fig 5.1: Signup Page

My Projects

Manage and view your generated architectures

[+ New Project](#)

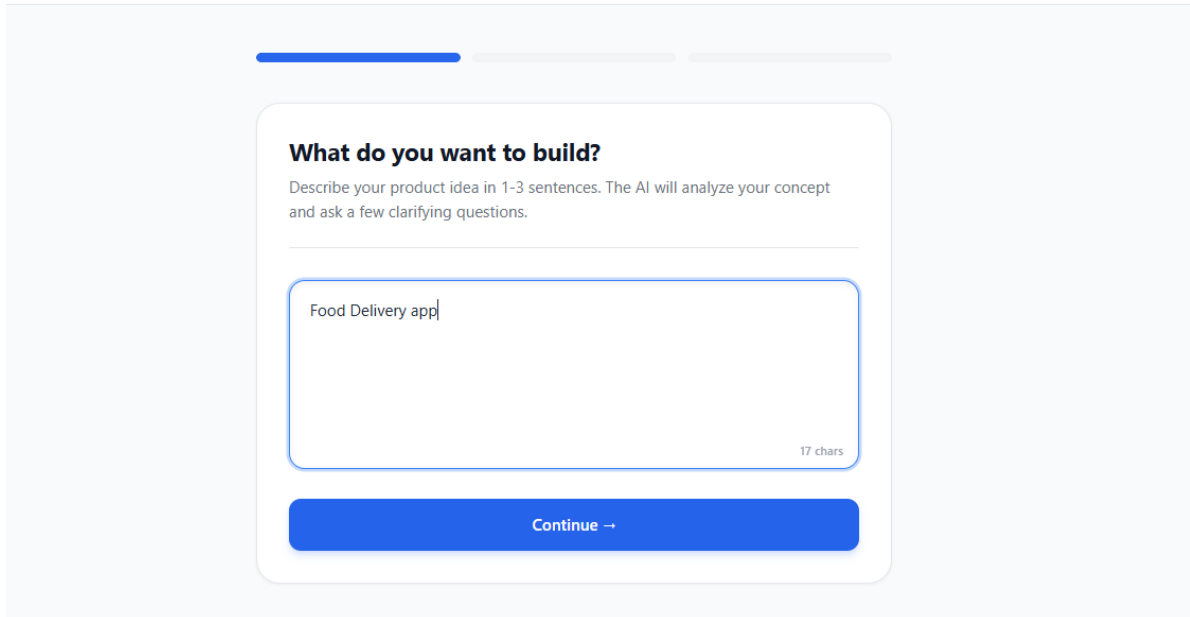


No projects yet

Create your first AI-generated architecture now.

[Create your first project →](#)

Fig 5.2: Dashboard on First Login



What do you want to build?

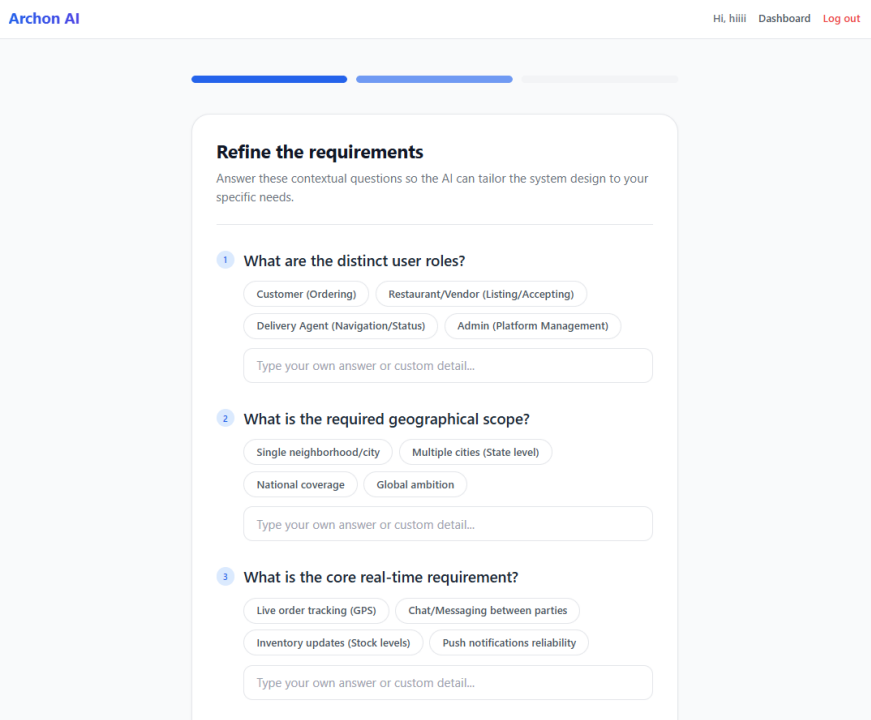
Describe your product idea in 1-3 sentences. The AI will analyze your concept and ask a few clarifying questions.

Food Delivery app

17 chars

[Continue →](#)

Fig 5.3: Project Ideation Wizard Step 1



Refine the requirements

Answer these contextual questions so the AI can tailor the system design to your specific needs.

- What are the distinct user roles?**

Type your own answer or custom detail...
- What is the required geographical scope?**

Type your own answer or custom detail...
- What is the core real-time requirement?**

Type your own answer or custom detail...

Fig 5.4: Project Ideation Wizard Step 2



Archon AI Hi, hiiii [Dashboard](#) [Log out](#)

Finalize your project

Give your project a name. We'll generate the full system design after this.

Project Name *

Helo Food

Short Description (optional)

Hi this is food delivery app

[← Back](#) [Auto-Architect System Design](#)

Fig 5.5: Project Ideation Wizard Step 3

Archon AI Hi, hello [Dashboard](#) [Log out](#)

[← Back to Dashboard](#) [Versions](#) [Export](#) [Refine with AI](#)

Advanced Notes

[Requirements](#) [Architecture](#) [APIs](#) [DB Schema](#) [Diagram](#)

Functional Requirements

- ✓ Users must be able to create and manage team accounts.
- ✓ Users must be able to create, read, update, and delete encrypted notes within a team context.
- ✓ The system must support rich text/Markdown editing for note content.
- ✓ The system must allow setting automated reminders on notes based on conditional logic (e.g., date, trigger).
- ✓ The system must enforce end-to-end encryption, ensuring the server never sees the plaintext content.

Non-Functional Requirements

- ✓ The system must maintain high data confidentiality, meeting E2EE standards.
- ✓ The system must support medium-scale concurrent reads/writes from team members.
- ✓ The web application must provide a responsive user experience for content editing and viewing.

Fig 5.6: Generated Project Requirements

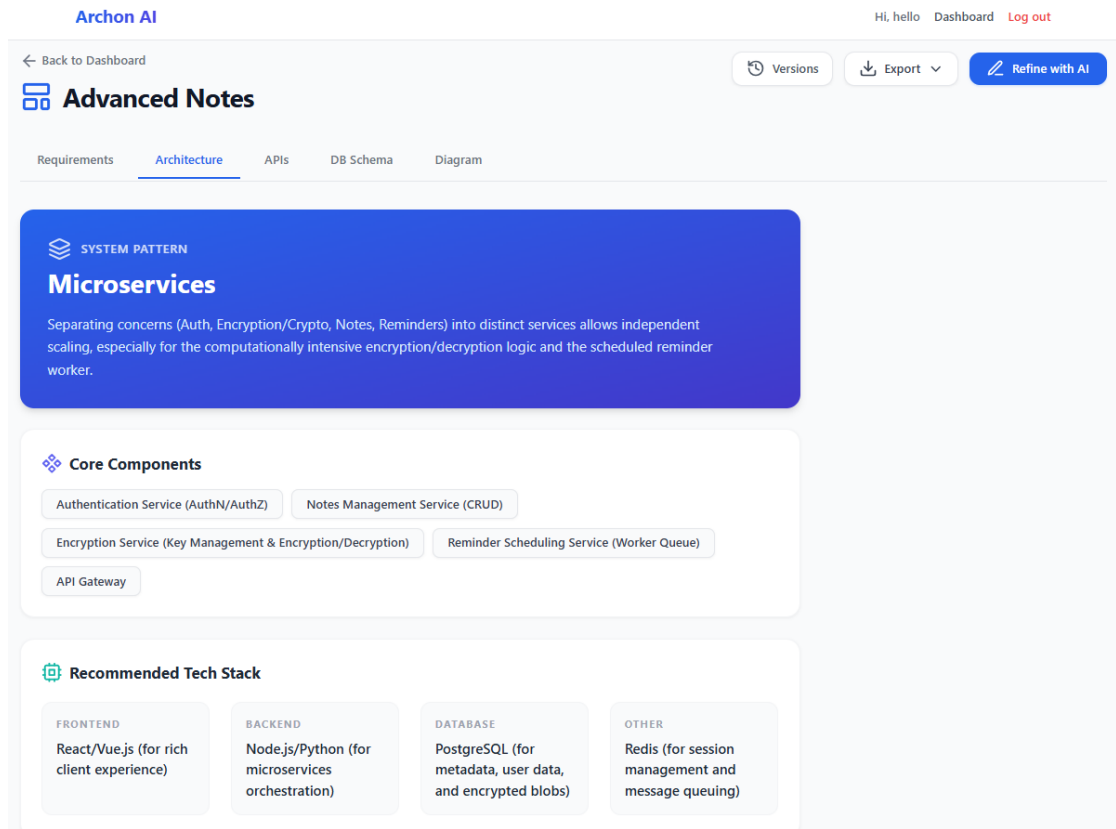


Fig 5.7: Generated Architecture for Project

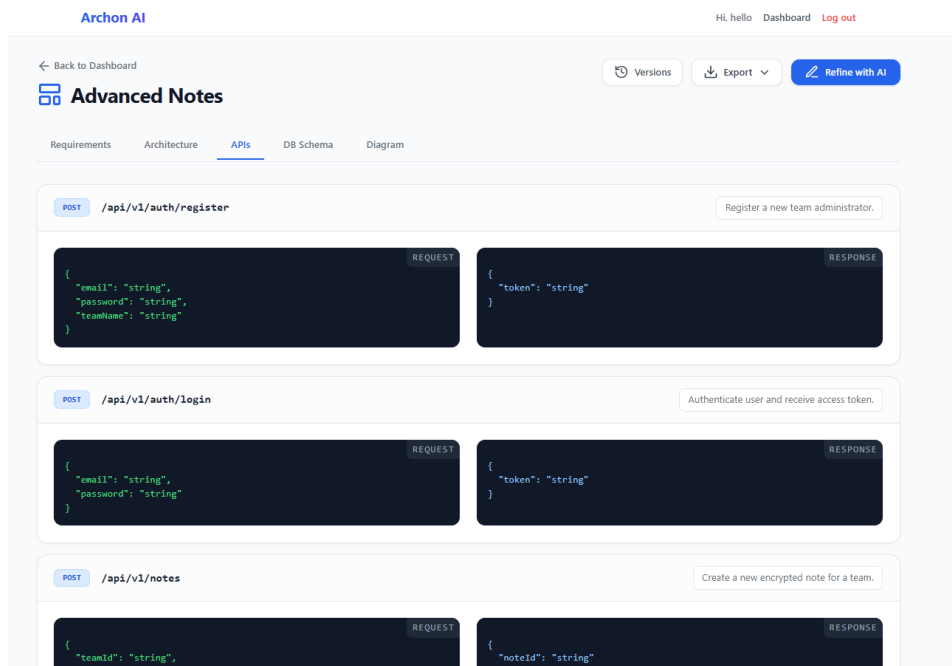


Fig 5.8: Generated API design



Archon AI Hi, hello [Dashboard](#) [Log out](#)

[← Back to Dashboard](#) [Versions](#) [Export](#) [Refine with AI](#)

Advanced Notes

Requirements Architecture APIs **DB Schema** Diagram

FIELD	TYPE
_id * PK	ObjectId
name *	String
ownerId * FK to users	ObjectId
encryptionKeyHandle Pointer to key management service	String

FIELD	TYPE
_id * PK	ObjectId
email * unique	String
passwordHash *	String
role * Admin/Member	Enum

FIELD	TYPE
_id * PK	ObjectId
teamId * FK to teams	ObjectId
title *	String
encryptedContent * Ciphertext blob	String
lastUpdated *	Date

FIELD	TYPE
_id * PK	ObjectId
noteId * FK to notes	ObjectId
teamId * FK to teams	ObjectId

Fig 5.9: Generated DB Schema

Archon AI Hi, hello [Dashboard](#) [Log out](#)

[← Back to Design](#)

Version History

v1

Design Iteration 1
[Export](#)

🕒 17/5/2026, 10:40:28 am

Fig 5.10: Version History of Project

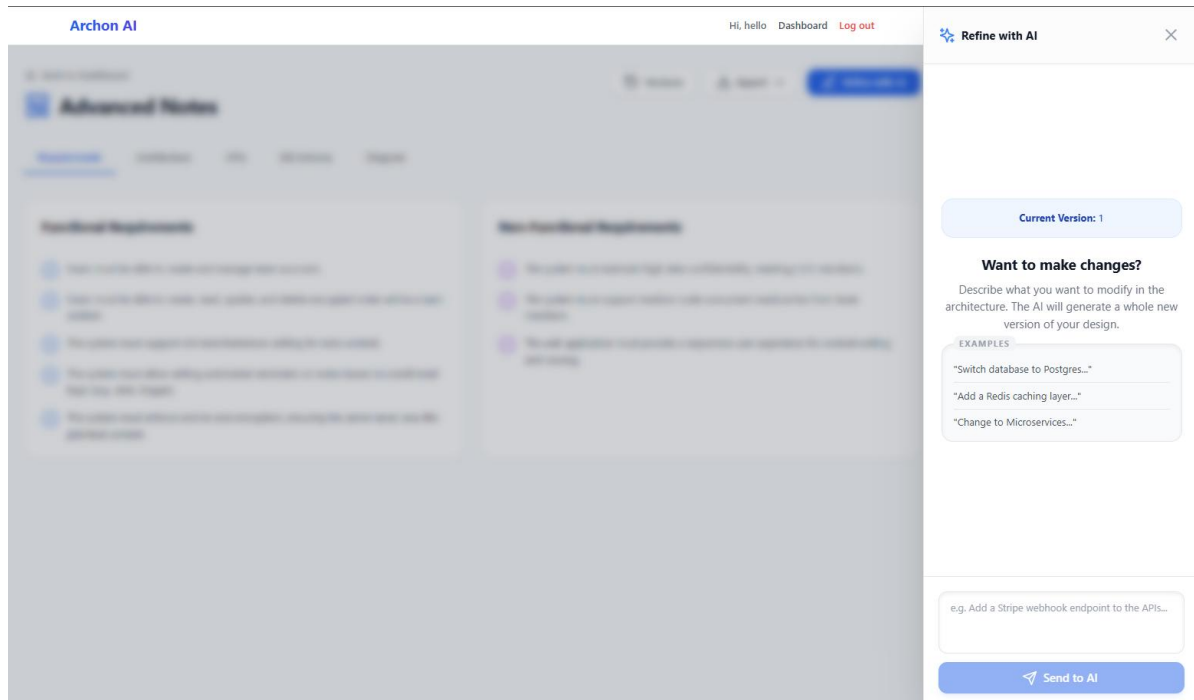


Fig 5.11: Side Panel to refine project with AI

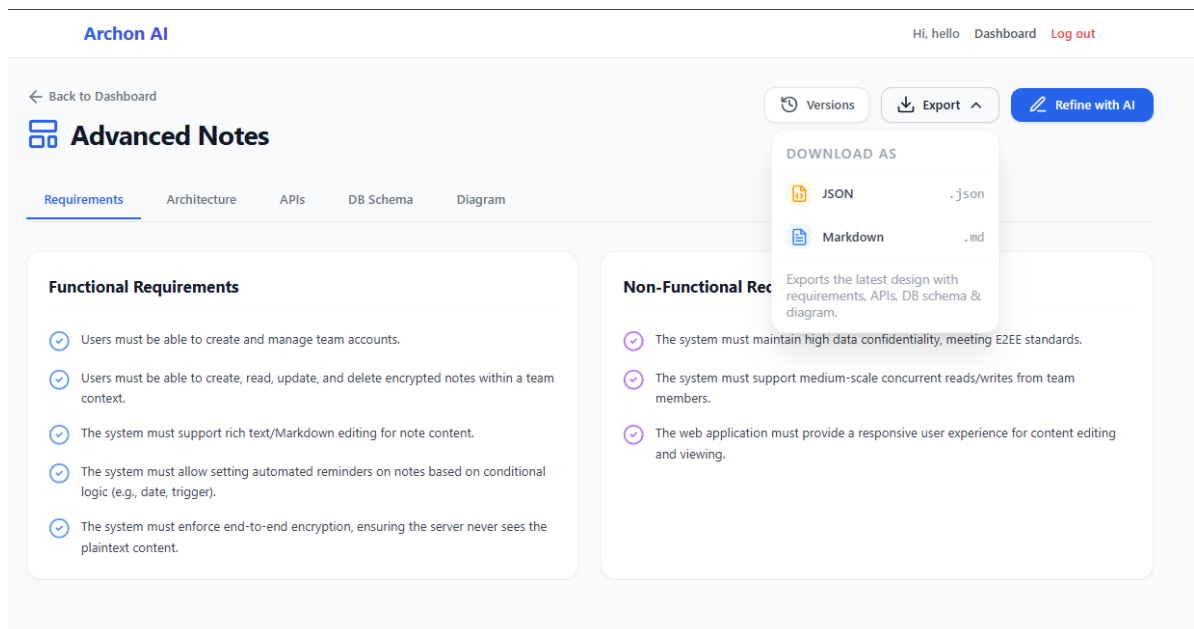


Fig 5.12: Export Options



```

1 # Advanced Notes – Architecture Design (v1)
2
3 **Generated:** Sun, 17 May 2026 05:10:28 GMT
4 **Status:** draft
5 **Design Version:** 1
6
7 ---
8
9 ## 1. Functional Requirements
10
11 1. Users must be able to create and manage team accounts.
12 2. Users must be able to create, read, update, and delete encrypted notes within a team context.
13 3. The system must support rich text/Markdown editing for note content.
14 4. The system must allow setting automated reminders on notes based on conditional logic (e.g., date, trigger).
15 5. The system must enforce end-to-end encryption, ensuring the server never sees the plaintext content.
16
17 ## 2. Non-Functional Requirements
18
19 1. The system must maintain high data confidentiality, meeting E2EE standards.
20 2. The system must support medium-scale concurrent reads/writes from team members.
21 3. The web application must provide a responsive user experience for content editing and viewing.
22
23 ## 3. Architecture
24
25 **Pattern:** Microservices
26 **Rationale:** Separating concerns (Auth, Encryption/Crypto, Notes, Reminders) into distinct services allows
independent scaling, especially for the computationally intensive encryption/decryption logic and the scheduled
reminder worker.
27
28 ### Tech Stack
29
30 | Layer | Technology |

```

Fig 5.13: Exported .md file for project

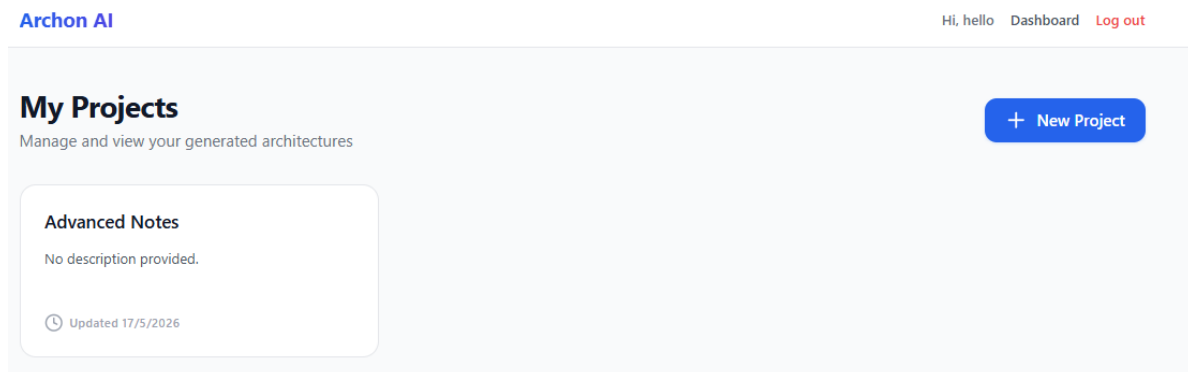


Fig 5.14: Project Dashboard

6. CODE IMPLEMENTATION AND DATABASE CONNECTIONS

6.1 Implementation Overview and Architecture Plan

The execution and deployment of Archon-AI are realized through a decoupled full-stack architecture using the MERN (MongoDB, Express.js, React.js, Node.js) framework. The system is built using strictly modular, component-driven coding standards to ensure vertical scalability, codebase maintainability, and optimal memory management during multi-tenant data operations. Rather than binding application services into a singular monolithic engine, every core feature—such as the real-time drawing canvas, the multi-turn conversational interface, or the assessment compiler—operates as an independent logical sub-module.

The frontend tier functions as a fluid Single Page Application (SPA) that communicates with the server entirely via asynchronous HTTP requests and persistent WebSocket protocols. The application tier runs a non-blocking Express.js routing engine designed to manage security validation middleware, file stream ingestion pipelines, and third-party AI gateway routing loops .

By decoupling presentation components from persistent datastore states, the platform achieves zero structural interference, allowing developers to upgrade backend models or add new database tables without rewriting client-side visualization logic.

6.2 Frontend Implementation Specifications

The client-side infrastructure of Archon-AI is engineered to deliver an intuitive, high-performance, and visually responsive interface for multi-tenant workspace management. Built using the core reconciliation principles of the React.js framework, the user interface updates views dynamically by executing targeted sub-tree rendering passes rather than forcing global browser updates.

Core Frontend Operational Deliverables

- **Multi-Tenant Command Dashboard:** Provides a centralized workspace view that dynamically tracks and presents user engagement trends, analytical file breakdowns, and project operation histories.
- **Asynchronous Multi-Turn AI Interface:** An interactive chat engine featuring auto-scrolling message bubbles, real-time input status bars, and dedicated target hooks to bind context questions to uploaded files.
- **High-Performance Canvas Workspace:** Built directly on top of the HTML5 Canvas API, this module tracks vector coordinates to synchronize multi-user drawing strokes across connected whiteboards in real time.
- **Responsive Layout Fluidity:** Styled with flexible grids and adaptive media breakpoints to ensure clean, readable dashboards across a wide variety of mobile, tablet, and high-resolution desktop screen profiles.
- **Client-Side Identity Validations:** Protects system routes by validating form layouts, verifying input syntax, and catching missing registration elements before dispatching login payloads to the API.

6.3 Backend Implementation Specifications

The backend engine of Archon-AI serves as the core orchestration center, handling client API calls, processing files, and managing downstream traffic to external generative AI nodes . Running inside a lightweight Node.js runtime, the server utilizes non-blocking I/O event loops to manage thousands of concurrent operational requests smoothly without dropping connections .

Core Backend Functional Areas

- **Cryptographic Identity Management:** Manages user authorization loops by intercepting incoming requests, evaluating secure JSON Web Tokens (JWT), and maintaining session security across protected routes.

- **Unstructured Asset Ingestion Pipeline:** Manages multi-part file uploads by securely receiving raw PDF streams, offloading the binary files to Cloudinary cloud repositories, and cleaning the characters for downstream text analysis.
- **Generative AI Gateway Orchestration:** Directs context tokens to the Google Gemini API. This module automatically builds system instructions and wraps context text inside precise prompt templates to ensure clean, structured outputs for summaries, explanations, and chatbot conversations .
- **Automated Assessment Compilation Engine:** Processes raw document text to dynamically generate interactive study tools, transforming text files into structured evaluation arrays and multi-sided flashcards .
- **Stateless Session Management:** Uses environment variables to securely store and inject API access tokens, private cluster URLs, and cryptographic cookie hashes, preventing data exposure.

6.4 Database Connections and State Persistence

The datastore tier of Archon-AI uses MongoDB to securely manage platform metadata, workspace state tracking profiles, and historical chat logs. Connection pipelines between the application server and the database cluster are managed via the Mongoose Object Data Modeling (ODM) framework, which enforces strict schema validations over flexible non-relational document trees.

Database Connection Responsibilities

- **Identity Ledger Retention:** Stores user registration profiles, encrypted login hashes, and security status records.
- **Workspace Metadata Management:** Tracks tenant allocation parameters, mapping workspace access lists and collaborator configurations directly to their respective owner IDs.
- **Document Asset Record Mapping:** Maintains live references to Cloudinary media files, matching file names, storage metrics, and extracted document text back to parent workspace entities.



- **Chat Context History Management:** Appends historical message sequences to specific user communication histories, preserving context for multi-turn AI chat conversations.
- **Evaluation Metric Aggregation:** Stores generated quiz arrays, tracking user option sets, correct answer references, and previous test performance metrics .

To maintain absolute stability under heavy data loads, the connection layer configures automatic pooling and connection time limits. The cluster monitors connections constantly, automatically routing around network drops or server lags to keep database read and write actions running smoothly across all active tenant workspaces.

6.5 System Data Flow and Architecture Diagrams

(Note: You can draw or model your visual charts based on the precise layout blueprints provided below.)

6.5.1 Comprehensive System Architecture Blueprint

This diagram maps out the hierarchical separation of tiers within Archon-AI, showing the flow of request-response communication loops across the entire platform ecosystem:

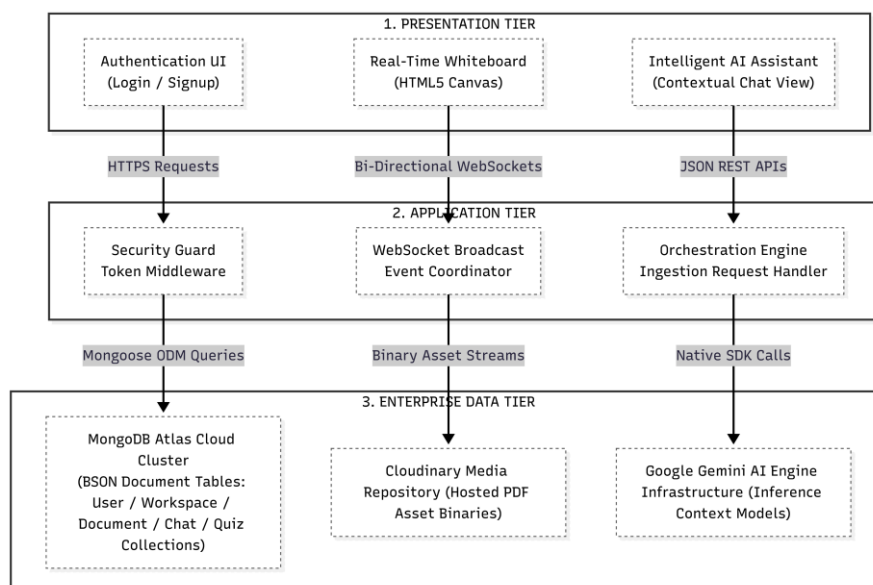


Fig 6.5.1: System Architecture



6.5.2 Document Vectorization and AI Orchestration Sequence

This data schematic details the sequential path data takes when a user uploads a PDF file to run context-guided AI generation tasks :

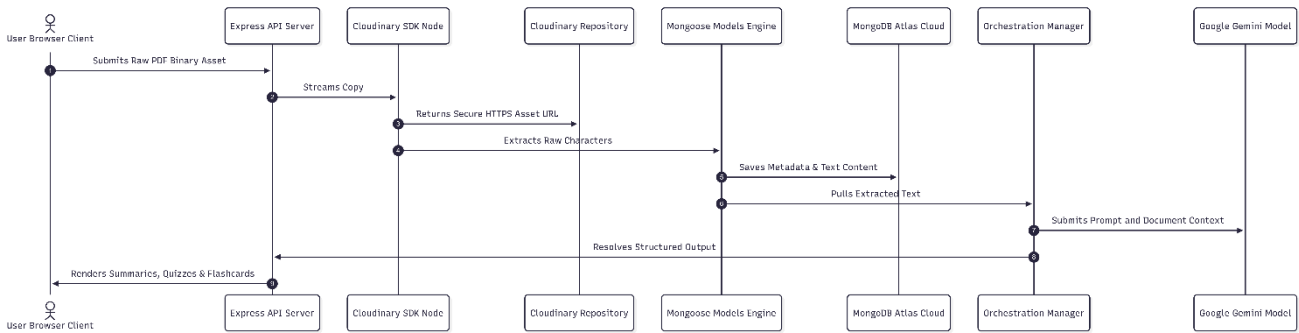


Fig 6.5.2: Sequence Diagram for AI orchestration

6.5.3 Real-Time Collaborative Whiteboard Synchronization Loop

This synchronization schematic outlines how drawing data transfers across team members to maintain identical canvas positions without lagging the user UI:

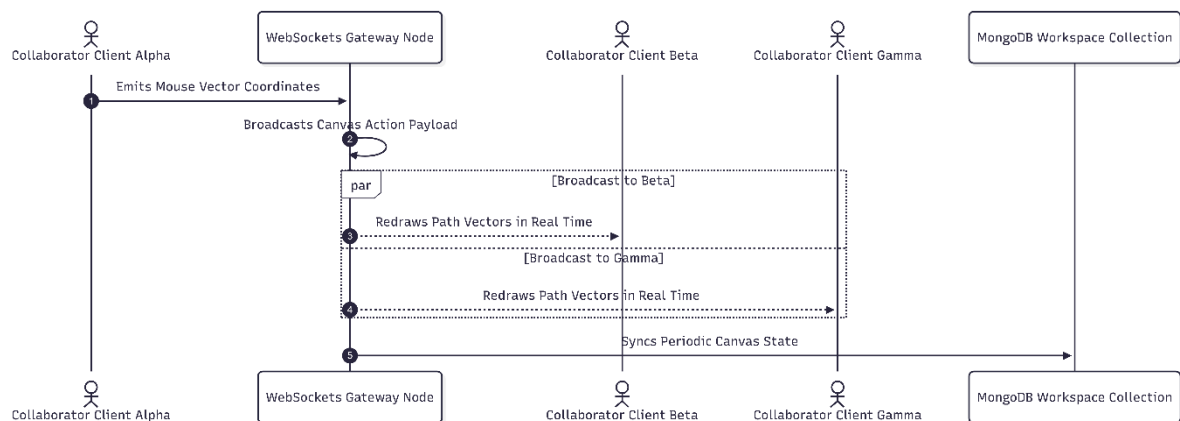


Fig 6.5.3: Sequence diagram for Whiteboard Sync

7. SYSTEM TESTING

7.1 Comprehensive Unit Testing Matrix

Unit testing verifies that the application's core logic blocks operate flawlessly in isolation before being integrated into broader system features . These tests isolate specific functions, apply targeted mock data, and confirm that our processing pipelines return the expected outputs.

Test Reference ID	Targeting Application Module	Test Operational Intent Summary	Input Variable Vector Parameters	Expected Code Assert Result Outcome	Verified Execution Pass / Fail State
UT-AUTH-01	Account Registration Service DOCX	Verify input data checker properties handle short passwords securely. DOCX	User Registration object: { username: "yuvraj", email: "y@university.edu", password: "123" } DOCX	Catch validation block and return a 400 Bad Request status error message. DOCX	PASSED
UT-AUTH-02	Account Registration Service DOCX	Confirm registration blocks duplicate email signups. DOCX+ 1	Registration attempt with an email address that already exists in the database.	Throw duplicate key violation exception error and block account generation. DOCX+ 1	PASSED
UT-AI-01	AI Request Controller Pipeline	Confirm error handlers safely catch missing project keys.	Request payload with an empty prompt block: { workspaceId: "ws_099", prompt: "" } DOCX	Intercept data validation error and drop execution before calling the Gemini API . DOCX	PASSED
UT-DOC-01	Document Management Engine DOCX	Validate system size monitors block massively bloated uploads.	Ingestion stream passing an oversized payload (>50 MB).	Terminate request immediately and return an explicit 413 Payload Too Large error code.	PASSED
UT-QUIZ-01	Assessment Compilation Hub DOCX	Validate code scores calculation metrics safely handle zero-value arrays. DOCX	Score array calculation request containing no input answer elements: [].	Gracefully fallback to calculate a zero score instead of crashing with a divide-by-zero math fault.	PASSED

Table 7.1: Unit Testing Results



7.2 Integration Testing

Integration testing ensures that our separate app layers—the client views, server API routers, cloud storage networks, and external generative models—communicate smoothly with one another without data drops.

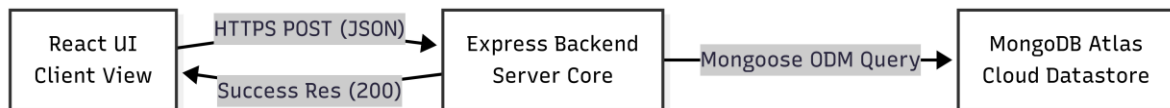


Fig 7.2.1: Integration Testing

7.3 Functional Testing

Functional testing evaluates the system against its real-world core operational requirements, ensuring that every end-user tool behaves as expected under normal usage conditions .

- **Collaborative Canvas Synchronization:** Tests the canvas processing loops across multiple browser connections. The test verifies that when user A creates vector shapes on the whiteboard, the drawing data is instantly synchronized across all other active workspace users without noticeable lagging .
- **Assessment Matrix Construction:** Tests the auto-generation pipelines for study tools. The test confirms that when a user runs the summary tool or generates interactive flashcards, the backend processes the raw file context and accurately returns fully populated, properly structured question layouts .
- **Workspace Management Flow:** Validates that workspace deletion and modification calls execute correctly without data leaks. The test checks that removing an asset successfully wipes its child documents, chat streams, and evaluation keys across all database tables.

7.4 UI / UX Testing

User Interface testing ensures that the interactive single-page application displays beautifully, loads quickly, and remains user-friendly across diverse browser platforms and screen resolutions .

- **Responsive Breakpoint Validations:** Verifies the layout adaptability using mobile and desktop viewing profiles. The test ensures that the document reader panel and chat tools collapse cleanly on narrower screens, allowing for fluid navigation without breaking interface components .
- **State Transition Verifications:** Traces data load indicators across the app during slow server connections. The test confirms that when the platform executes long tasks (such as a large PDF parse or quiz compilation), clear loading bars or spin animations appear to keep the UI interactive.

7.5 Database Testing

Database testing evaluates transaction accuracy, structural security rules, and data persistence rates across our cluster collections .

- **Data Lifecycle Validation:** Confirms that CRUD operations process correctly across our collections without distorting character symbols or breaking linked schema connections.
- **Concurrent Request Performance:** Measures datastore response latencies during simultaneous user bursts. The test proves that index locks remain stable, preventing duplicate email signups or split-tenant data leaks during high-traffic intervals.

7.6 Security and Performance Testing

7.6.1 Security Testing Parameters

- **Endpoint Protection Assertions:** Proves that directly calling API routes without passing structural credentials triggers an immediate 401 Unauthorized block from our middleware.

- **Input Injection Defenses:** Validates that trying to pass malicious script tags or database query symbols into our chat forms triggers safe syntax handling, protecting our backend routers from injection style attacks.

7.6.2 Performance Testing Parameters

- **Platform Loading Benchmarks:** Measures client app startup and rendering speeds under standard connections, showing a rapid time-to-interactive metric of <1.8seconds.
- **API Round-Trip Metrics:** Confirms that standard backend communication tasks (excluding external generative model streams) complete their execution loops within a crisp window of <120ms.

8. LIMITATIONS

While Archon-AI provides a highly secure, multi-tenant workspace experience, specific technical constraints exist in the current implementation that define boundaries for production rollouts :

- **Mandatory Cloud Network Connections:** Because the platform relies entirely on remote infrastructure for cluster hosting (MongoDB Atlas) and real-time inference handling (Google Gemini), the platform cannot run offline . If internet connections drop, users cannot access document analysis, chat tools, or interactive quiz generation features .
- **Document Architecture Restrictions:** The file processing pipeline is strictly optimized to read text from standard PDF documents. The system does not natively support rich media uploads like raw .docx text formats, structural .pptx slides, or unparsed image attachments.
- **Generative Token Limits:** When processing exceptionally long documents, the background text context can hit model token limit caps. This ceiling can lead to truncation errors, causing the model to miss data found at the very end of massive documents .
- **Basic Processing Latency:** The processing runtime for large documents scales with file size. Generating automated summaries or comprehensive quiz decks from large files can introduce variable delivery lags based on current traffic at the cloud API endpoints.
- **Basic Collaborative Features:** While the real-time whiteboard allows for simultaneous drawing, it lacks granular workspace tracking features, multi-tier collaborator permission settings, or an embedded team audio link.

9. CONCLUSION

Archon-AI successfully demonstrates how a multi-tenant workspace platform can combine daily personal organization utilities with advanced generative models inside a cohesive system . Developed using the MERN web stack (MongoDB, Express, React, Node.js), the platform addresses the challenge of application fragmentation and context-switching overhead.

By embedding the Google Gemini API directly into the core controllers, the system changes how users manage files, transforming static text documents into interactive repositories capable of generating real-time summaries, custom chat streams, and responsive study tools .

The platform's robust schema rules and scalable MVC backend design ensure secure data tracking and reliable performance across distinct user tenants . Ultimately, Archon-AI provides a highly extensible foundation for building modern software tools that successfully put artificial intelligence directly into real-time collaborative workspaces.

10. FUTURE SCOPE

The underlying architecture of Archon-AI is built to support ongoing tool upgrades and system scaling loops :

- **Vector Database Integration (RAG):** Transitioning from standard raw text extraction to a complete Retrieval-Augmented Generation (RAG) pipeline . Integrating dedicated vector search databases (such as Pinecone or MongoDB Atlas Vector Search) will let the system index vast document libraries, performing rapid semantic search tasks across huge document networks .
- **Advanced Collaborative Whiteboard Upgrades:** Expanding the vector canvas using advanced web socket networks to support real-time user cursor tracking, inline text annotations, and shape manipulation mechanics .
- **Multi-Format Ingestion Pipelines:** Upgrading our ingestion engine to natively process additional file formats such as raw Word documents (.docx), presentation decks (.pptx), and markdown files, supplemented by Optical Character Recognition (OCR) tools to parse text from uploaded images or handwritten notes .
- **Granular Multi-Tenant Security Controls:** Developing role-based access control layers (RBAC) to allow workspace owners to assign specific permission tiers (such as Viewer, Editor, or Administrator) to different collaborators .
- **Cross-Platform Mobile Applications:** Porting the single-page web view into cross-platform mobile environments using React Native, allowing users to safely access their workspaces, interact with documents, and review study decks on iOS and Android devices .

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- **Google Gemini Foundational AI Models:** Prompt Engineering and Context Token Integration Mappings (<https://ai.google.dev/gemini-api/docs>)
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Engineering Communities & Technical Resource Libraries

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- **GeeksforGeeks Developer Portal:** Structural overviews of full-stack MERN state management patterns
- **Stack Overflow Technical Knowledge Base:** Best practices for handling streaming API responses and managing concurrent database queries